log1

In computational neuroscience, a common approach to modeling brain activity at rest involves fitting a system of linear stochastic differential equations to neuroimaging data. Once the model is fitted, it gives rise to a stochastic system of the form:

$$\dot{x}(t) = Ax(t) + w(t)$$

where A is the state-transition (effective connectivity) matrix, and $w(t)$ is a zero-mean white noise process with covariance $\Sigma_w = \mathbb{E}[w(t)w^T(t)]$. We assume $\Sigma_w \succ 0$ (positive definite) so that the noise perturbs the system in all directions, and A is Hurwitz stable (all eigenvalues have negative real parts), ensuring the existence of a positive definite steady-state or stationary covariance matrix:

$$\Sigma = \lim_{t\to\infty} \mathbb{E}\left[x(t)x(t)^T\right] \succ 0$$

which can be computed as the solution of the algebraic Lyapunov equation [1]:

$$A\Sigma + \Sigma A^T = -\Sigma_w$$

In linear DCM, Σ is the steady-state (zero-lag) covariance that reflects the instantaneous mutual influences between different brain regions (functional connectivity) [2].

The product $A\Sigma$ (differential covariance) represents the application of the dynamics defined by A to the steady-state covariance structure Σ, describing how the instantaneous patterns of neural activity evolve under the influence of the effective connectivity. We can decompose this term into symmetric and skew-symmetric parts:

$$A\Sigma = \underbrace{\frac{1}{2}(A\Sigma + \Sigma A^T)}_{\text{Symmetric}} + \underbrace{\frac{1}{2}(A\Sigma - \Sigma A^T)}_{\text{Skew-Symmetric}}$$

With $S = \frac{1}{2}\left(A\Sigma - \Sigma A^T\right)$, using the Lyapunov equation:

$$\begin{aligned} &A\Sigma + \Sigma A^T = -\Sigma_w \\ &\implies 2(A\Sigma - S) = -\Sigma_w \\ &\implies A\Sigma = -\tfrac{1}{2}\Sigma_w + S \end{aligned}$$

where:

- **Differential Auto-Covariance (dA-Cov)** $-\frac{1}{2}\Sigma_w$: Diagonal matrix beacuse of uncorrelated noise assumption [3] IT MEANS THAT INTRINSIC SELF CHANGE OF ACTIVITY IS DUE TO ONLY RANDOM INJECTION. Governs the dissipative dynamics. Reflects how the injected noise is dissipated by the system. Corresponds to a gradient flow, meaning trajectories are driven directly towards equilibrium.
- Differential Cross-Covariance (dC-Cov) S: Represents rotational or solenoidal flows in state space—directions of movement that conserve energy and do not align with gradients. Encodes for the irreversibility or asymmetry—which region drives which, over time. Critical in understanding the non-equilibrium, directed flow of information in the brain, beyond what static FC measures provide.

In the context of brain dynamics, while functional connectivity Σ gives an undirected, symmetric picture of how brain regions co-activate at zero-lag, the effective connectivity (and in particular its skew-symmetric component S) reveals the directionality of the interactions. Empirically this allows one to identify which regions act more as sources and which as sinks, an insight into the hierarchical structure of brain networks.

Given that the covariance matrix Σ evolves with dynamics A, namely $A\Sigma$, its solution for a time lag τ is

$$P(\tau) = e^{A\tau}P(0)$$

where $P(0) = \Sigma$ [4]. Any two nodes i and j in the parcellation sit some physical distance r_{ij} apart. The (i, j) entry of the covariance matrix $P(\tau)$ represents how much the activity in i co-varies with the activity in j at lag τ. In a homogeneous, isotropic system (one that looks the same everywhere and in every direction), it makes sense to pool together all pairs of nodes that happen to be separated by the same distance r. On average, that gives the function

$$B(r;\tau) = \langle P_{ij}(\tau)\rangle_{\{(i,j):d_{ij}\approx r\}}$$

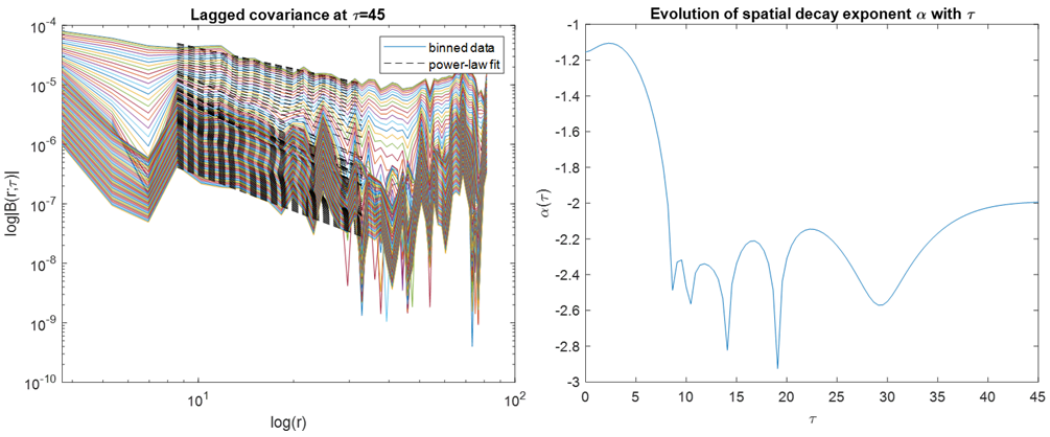
where d_{ij} is the physical distance between parcel i and parcel j. If $B(r)$ follows a power-law $B(r) \propto r^{\alpha}$, then

$$\log B(r) = \alpha \log r + c$$

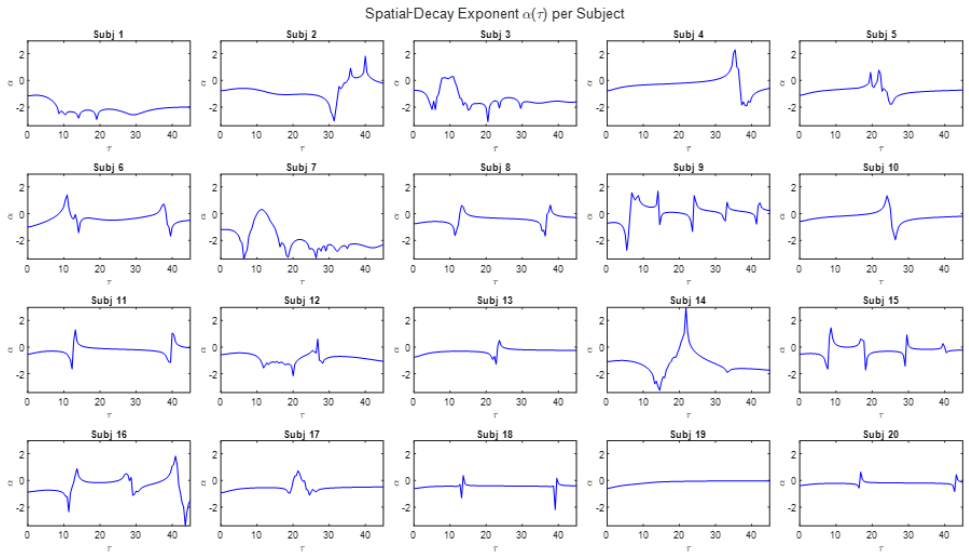
In turbulence theory (Kolmogorov's structure functions) you look for exactly this kind of scaling in an inertial subrange of intermediate distances—neither so small that you're dominated by noise, nor so large that everything decouples [5]. The slope α of a fitted power law

$$B(r;\tau) \propto r^{\alpha(\tau)}$$

is a measure of how quickly the covariance falls off with distance at each time-lag τ. The closer α is to zero, the slower correlations decay. It represents how fast the cortex transmits fluctuations and how long it retains memory of those fluctuations before they decorrelate. It represents how far in space and time the spontaneous activity of the brain remains correlated.



Test 1 (100 samples at maximum lag $\tau = 45\,\mathrm{s}$) exhibits oscillatory dips, which could come either from noisy fitting or actual dynamical transition in the effective connectivity. A deeper dip means stronger local segregation (correlation decays faster in space—regions are more independent—regions exhibit stronger local segregation). When α becomes less negative, spatial spead increases again, which could reflect long-range propagation effects—a wave-like spread damped at certain scales.



Some subjects show clear oscillatory bumbs, pointing to resonant or wave-like network modes coming and going at those time-scales. Others are almost flat, these brains either don't have a strong spatial-scale preference at any particular delay, or else the dynamics are too noisy (or too weak) to pull out a reliable exponent. Some sit in between, suggesting a weakly underdampened mode.

Clustering subjects on the basis of shape and need for robust summary metrics.